



Agent-Based Modeling Under Climate Uncertainties: The Coastal Resilience (CRes-ABM) Model

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Abstract

The Coastal Resilience Agent-Based Model (CRes-ABM) is introduced, which was created to evaluate the resilience of coastal areas to the effects of climate change, with a particular emphasis on sea-level rise and extreme weather events. The aim is to provide information for developing policies and making decisions about boosting coastal resilience. CRes-ABM simulates stakeholder interactions and adaptation responses within coastal communities by integrating agent-based modeling techniques, geospatial data, and climate change scenarios. On the basis of the CResI index and the effect of adaptation strategies (CResI_adapt), CRes-ABM produces vulnerability-resilience maps. A variety of parameters are used to monitor agent decisions throughout each simulation. We compare the outcomes using historical data (1976–2005) and two custom climate change scenarios (2041–2100). A thorough framework for comprehending and addressing issues related to coastal resilience is provided by CRes-ABM, which makes it easier to make evidence-based decisions about successful adaptation measures. The structure of CRes-ABM allows it to be applied to various coastal regions worldwide, enabling the evaluation of resilience across different geographic, climatic, and socio-economic contexts.

Keywords Climate change adaptation · Coastal resilience index · Sea-level rise · Coastal resilience assessment · Coastal flooding · Agent-based modeling

Highlights

We present CRes-ABM, a new agent-based model for climate change adaptation.
CRes-ABM uses the previously developed coastal resilience index (CResI).
CRes-ABM integrates cell, household, and governmental agents.
CRes-ABM assesses climate change impacts considering environmental and socio-economic scenarios.
CRes-ABM advantages include computational speed, processing using the GAMA platform, and constant visualization of results.

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Introduction

Erosion and floods pose a serious hazard to coastal regions globally, affecting both public and private infrastructure. This covers residences, hotels, airports, power plants, and facilities for treating wastewater (Giannakidou et al. 2020; Griggs and Reguero 2021; Stamou et al. 2024). Meanwhile, changes in land cover increase surface temperatures in urban areas (Mazumder et al. 2024). It is becoming more and more necessary to address these issues as coastal cities and communities have to deal with expensive, time-consuming, and restricted options for environmental reviews and permits (Chowdhury et al. 2019). Infrastructure built in

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the twenty-first century faces serious threats from erosion, flooding, strong storms, and sea-level rise.

Coastal regions around the world face the urgent threat of erosion and flooding, which impacts both private and public infrastructure. This includes homes, hotels, airports, power plants, and wastewater treatment facilities (Giannakidou et al. 2020; Griggs and Reguero 2021; Stamou et al. 2024). Although higher flood susceptibility does not always imply significant flood risk (Dey et al. 2024), addressing these challenges is increasingly urgent, as coastal communities and cities must navigate through limited solutions and complex, costly, and time-consuming permitting and environmental review processes (Chowdhury et al. 2019).

Infrastructure built in the twenty-first century faces serious threats from erosion, flooding, strong storms, and sea-level rise. Altering characteristics of the physical and socioeconomic environment has impact on the livability of urban and suburban areas (Najafi et al. 2024). Proactive planning and reevaluation of development and resource management in coastal regions are essential today (Gaur and Simonovic 2015), even if the most severe effects of sea-level rise may not be felt for some time (Hinkel et al. 2013; Afentoulis et al. 2017; Beck et al. 2018; Reguero et al. 2021).

In this paper, the creation of an agent-based model to evaluate coastal metropolitan areas' resistance to climate change is presented, after creating tailored climate change scenarios for Greece, derived based on the most recent climate models (CMIP6) included in the IPCC's sixth assessment report (AR6) (Calvin et al. 2023). The purpose of the proposed Coastal Resilience Agent-Based Model (CRes-ABM) is to model and evaluate how resilient coastal regions are to the effects of climate change, including rising sea levels and harsh weather. It examines how stakeholders in coastal communities interact, adapt, and respond to environmental changes using a combination of agent-based modeling approaches, climate change scenarios, and geo-spatial data. Using the recently developed coastal resilience index (CResI) (Roukounis et al. 2023), the objective is to evaluate the resilience of coastal metropolitan areas as well as their susceptibility to floods and sea-level rise. The application area of the model is the wider area of Vari-Voula-Vouliagmeni, located in the southwest part of the Athens Metropolitan area.

The CResI is an innovative tool that includes 19 parameters grouped into six categories: geomorphology, flooding, wave exposure, land use, socioeconomic factors, and infrastructure/functionality (Roukounis et al. 2023). By identifying who is susceptible rather than merely what is vulnerable, it employs GIS tools to provide a thorough perspective of interactions between humans and their environment (Roukounis and Tsihrantzis 2022). The assessment of the effects of rising sea levels and more frequent

coastal flooding as a result of climate change on the sustainability and resilience of urban coastal communities is one of CRes-ABM's goals. In order to forecast the ramifications of sea-level rise and flooding patterns for metropolitan areas, this model painstakingly explores a variety of scenarios.

In addition to evaluating the immediate effects of these environmental changes, CRes-ABM highlights the efficacy of various adaptation and mitigation tactics at the individual (household) and governmental levels. A comprehensive analysis of the various ways that governments and society could respond to increasing climate risks is provided by the CRes-ABM output.

Literature Review

Agent-Based Modeling

The assessment of climate change impacts and the formulation of adaptation and mitigation strategies have traditionally relied on the use of integrated assessment models (IAMs). These models contain a basic computable general equilibrium framework with a simplified carbon cycle module (Nordhaus 1993; Tol 2001; Bosetti et al. 2013; Golosov et al. 2014). However, IAMs have encountered growing criticism in the literature due to their reliance on oversimplified assumptions and possible lack of transparency around model structures (Gambhir et al. 2019).

Agent-based modeling (ABM) has its roots in the early developments of cellular automata (CA) (Von Neumann and Burks 1966), a theoretical framework comprising an array of cells that form a discrete spatial two-dimensional domain. In this construct, each cell processes inputs related to specific characteristics, leading to different "states" as outcomes (Adamatzky 2010). While cellular automata may appear simple, they are able to execute diverse computations and simulate real-world processes (Crooks et al. 2018, 2019). ABMs follow the same logic, to create dynamic modeling tools which have the agents as central entities to simulate human and natural behaviors, with interactive feedback mechanisms.

In recent years, a new generation of agent-based models has been developed to examine possible connections among economic growth, energy dynamics, and climate change at regional, national, and global levels. A growing number of studies are conducted in this field by Gerst et al. (2013); Hasselmann and Kovalevsky, (2013); Inkoom et al. (2017); Grumbly et al. (2019); Guimaraes et al. (2021); Nava-Guerrero et al. (2021); Fouladvand (2022, 2024); Madler et al. (2023); and Ribeiro-Rodrigues and Bortoleto (2024).

Agents in ABMs can be defined as self-governing entities capable of making decisions based on various criteria (Simmonds et al. 2020). Agents in urban resilience studies represent a diverse range of actors, such as investors, representatives from the public and private sectors (including governmental ministries, municipalities, and port authorities), policymakers, relevant businesses, and consumers. ABMs view the real world as a complex and ever-evolving system (Filatova et al. 2009; Kirman 2016; Dosi and Roventini 2019; Cincotti et al. 2022), and as a result, they serve as a useful tool for modeling complex and dynamic systems. For this, they simulate the interactions among various types of agents, illustrating how these interactions can influence both the agents themselves and their surrounding environments. Meanwhile, technological advancements led to increase in data availability at macro- and micro-scale level, as well as computational power (Abar et al. 2017). As a result, constant socio-environmental changes required advanced modeling techniques, with ABMs contributing significantly in this area (Savin et al. 2023).

The main challenge for ecological modeling was to find more realistic representations of human behavior beyond the simplifying assumptions of the “rational actor” model, which assumes that individuals optimize their self-interest. Instead, there is a growing interest in agent-based modeling (ABM) applications to capture the complexities of human behavior (Egli et al. 2019; Gotts et al. 2019; Wijermans et al. 2020; Taghikhah et al. 2021). More and more applications are developed in different sectors, including transportation (Milne et al. 2022; Calabrò et al. 2023; Diallo et al. 2023), marine and port services (Vaněk et al. 2013; Fransen and Davydenko 2021; Cotfas et al. 2023), energy use (Muelder and Filatova 2018; Tian and Chang 2020; Mussawar et al. 2023), and consumer behavior (Ernst and Briegel 2017; Scalco et al. 2019; Koide et al. 2023). In the field of climate change, integrated assessment ABMs are also a useful tool for planners and decision makers (Di Noia 2022).

Various adaptation measures for climate change in coastal areas have been explored in the literature, including coastal defense (Nikolic and Simonovic 2015; Han et al. 2020; Mills et al. 2021; Laatabi et al. 2022; Santamarta et al. 2024) and management (Zhang et al. 2015; Abebe et al. 2019). The literature emphasizes the importance of context-specific approaches to coastal adaptation (Lawyer et al. 2023) and provides valuable guidance for policymakers and practitioners. For the purpose of this research, we used standard databases and searched for keywords related to climate change, adaptation, agent-based modeling, floods, flood-proofing, coast, managed retreat, household, green infrastructure, and policy. Relevant publications were filtered by title and abstract contents.

Agent-Based Modeling Platforms Using GIS Techniques

In the first years of the twenty-first century, early general-purpose agent-based modeling (ABM) frameworks started growing in the literature. While these frameworks often lacked direct support for geographical data, some introduced extensions or plugins offering limited capabilities for loading, transforming, and visualizing such data. For instance, Swarm (Minar et al. 1996) included the Kenge library (Box 2001, 2002), enabling the incorporation of GIS data in the form of grid cells, but has been inactive since 2011. NetLogo (Wilensky 1999), one of the pioneering ABM frameworks, ranks (CoMSES 2022) as the most popular framework for developing agent-based models shared online and utilized in publications. NetLogo provides a GIS extension to handle geographical data. However, users often need to manually convert and map loaded GIS data into NetLogo’s discretized space, represented as patches. Additionally, recreating polylines requires generating links, while shortest path calculations are only available through a separate network extension (Crooks et al. 2019).

State-of-the-art ABMs contain geospatial support, to cover the increasing need to integrate location-specific data for the agent’s environment. AnyLogic (Borshchev 2014), a modern framework, features built-in GIS support. MASON (Cioffi et al. 2014), another widely-used framework, incorporates GIS functionalities through its GeoMason (Sullivan et al. 2010) extension. However, GIS operations such as processing, editing, and analyzing geographical data typically require external tools like ArcGIS or QGIS. Repast (North et al. 2013) offers similar GIS capabilities. Recent development efforts within Repast have concentrated on large-scale distributed simulations in C++ (Repast HPC; North et al. 2013) and its Python interface (Repast4Py; Collier and Ozik 2022). GAMA (Taillandier et al. 2019), a more recent ABM framework with built-in GIS support, necessitates the use of its proprietary language, GAML (GAma Modeling Language), for model development.

Furthermore, modern programming languages led in further development of GIS-supported ABMs, e.g., in Python (Mesa (Kazil et al. 2020) and AgentPy (Foramitti 2021)) and Javascript (AgentScript; Densmore 2012).

Materials and Methods

Study Area

The study area is the greater area of Vari-Voula-Vouliagmeni. It is located at the south coastal waterfront of the continuous Athens Urban Area, Greece. Within its limits, there are urban, suburban, and rural areas, with different

types of coastlines (natural coastline, beaches, quay walls, small harbors, and marinas). Due to its natural beauty, this area also has high touristic interest and high property values (ELSTAT 2023). However, previous work (Roukounis et al. 2023) showed that the area is exposed to the impacts of climate change, due to its geomorphology and wave exposure. The selected area, with the point locations of the household agents, is shown in Fig. 1 and in Supplementary Material (SM) Fig. SM1.

The model environment is in a detailed grid system with a resolution of 100×100 m. This scale provides well-balanced computational efficiency, allowing for realistic representation of the coastal environment while maintaining manageable data volumes and processing times. Each grid cell is an agent representing a specific section of the coastal area. Agent characteristics include all the environmental and socio-economic attributes derived from the CResI.

The temporal scale of the CRes-ABM is set to monthly time steps, starting from 2041 until 2100, due to data availability for this period. The time-series contain data on mean wave height, maximum wave height, wave direction, and significant wave period. We compare the results of the future projections with data from the current conditions (period 1976–2005). Monthly time steps allow the model to capture seasonal variations in environmental conditions, such as sea-level rise and extreme weather events, as well as changes in socio-economic factors like population density and urban development.

Meanwhile, it is more straightforward to observe the effects of agent decision within the model, on both the agents and the environment under different scenarios. The authors took into account the lack of government adaptation planning and the lack of public awareness.

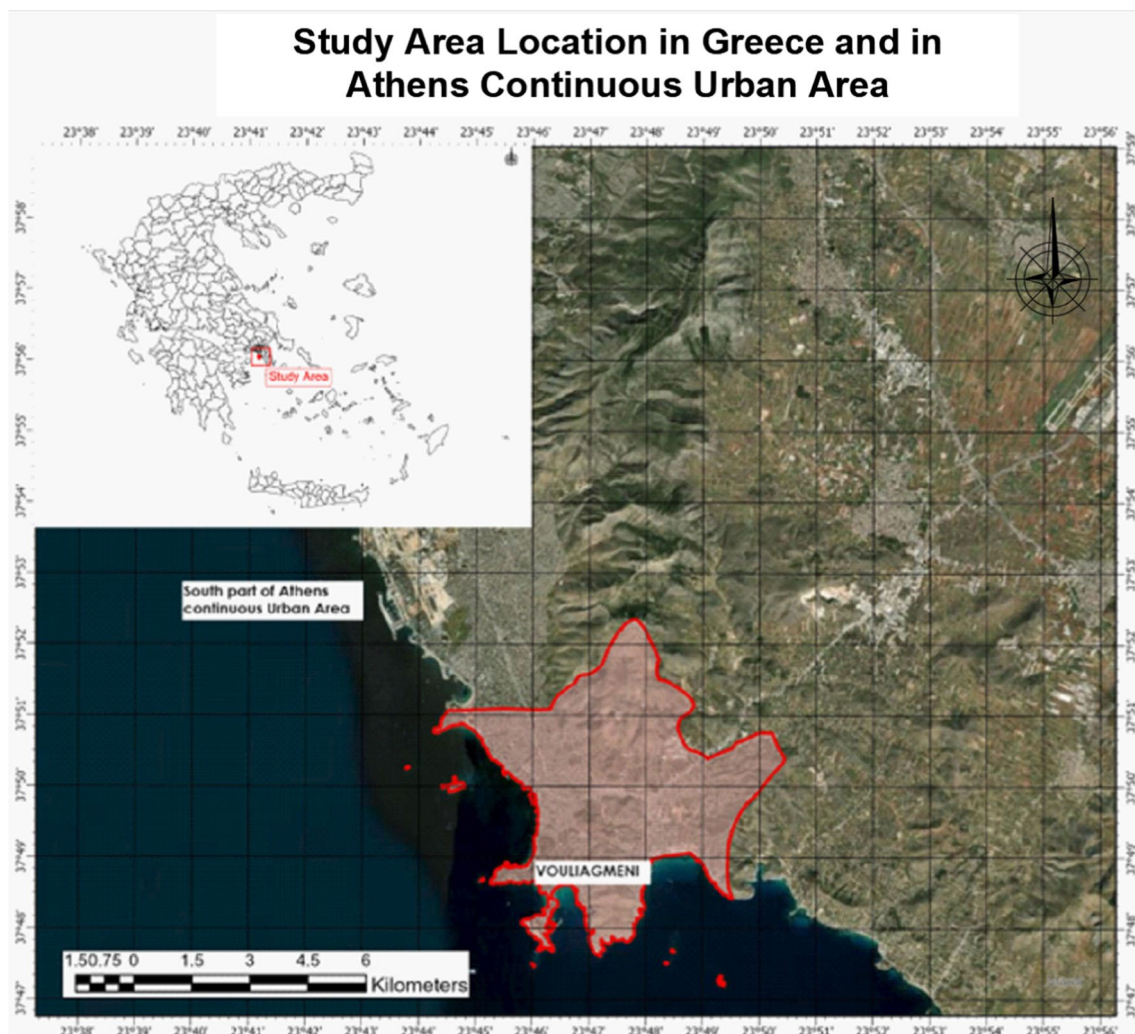


Fig.1 Map of the study area, Vari-Voula-Vouliagmeni, located in the greater Metropolitan Area of Athens, Greece. The insert map indicates location of the study area in Greece

A National Plan for Adaptation to Climate Change (NPACC) serves as a strategic framework to enhance resilience to climate impacts and guide long-term, sustainable decision-making. It aims to systematize adaptation efforts, integrate them into regional and sectoral development plans, and strengthen societal adaptive capacity. Sector-specific actions target agriculture, forestry, biodiversity, water resources, tourism, energy, infrastructure, and cultural heritage, emphasizing knowledge acquisition, sustainable resource management, risk mitigation, and stakeholder education. The plan also prioritizes the monitoring, evaluation, and continuous updating of adaptation strategies to address vulnerabilities and foster a climate-resilient economy and society.

Data Sources

The data sources of the CResI attributes are described in Roukounis et al. (2023). The dataset from Copernicus Service (Caires and Yan 2022) was used for the purpose of this research, for the future projections of wave heights. This dataset provides a comprehensive analysis of wave climate indicators, focusing on ocean surface wave parameters across a broad European domain.

The wave climate is characterized using key wave spectral parameters, such as significant wave height and peak wave period. To evaluate the impact of climate change on the ocean's surface wave field, the ECMWF wave model (ECMWF 2020) is employed under three distinct climate scenarios: the current or "historical" climate scenario, and two representative concentration pathway (RCP) scenarios. The wave data are used later in the development of the tailored scenarios for Greece (GR-SSPxRCP), described in the following section.

Data on SLR have been extracted from the NASA Sea Level Projection Tool (NASA 2022), a sophisticated digital resource designed to facilitate user access to sea level projection data derived from the AR6 (Calvin et al. 2023).

Moreover, for information regarding population change and other economic data, the database of the International Institute for Applied Systems Analysis (IIASA) is used (Riahi et al. 2017). Projections on income inequality for the shared socioeconomic pathways are derived from Rao et al. (2019). Economic data regarding GDP per capita in Greece and wealth distribution among households are obtained from the Hellenic Statistical Authority (ELSTAT) (2023) and the Institute of Small Businesses of the Hellenic Confederation of Professionals Craftsmen and Merchants (GSEVEE) (2022). The annual number of storms is a stochastic variable, based on historic data (HCMR 2022; Hochman et al. 2022).

Information on policy implications were derived from ADENS SA (2021). The Regional Plan for Adaptation to Climate Change (RPACC) assesses climate risks, the

interplay between climate and socioeconomic factors, the influence of adaptation actions on risk levels, and the prioritization of actions to address threats. Additionally, it examines the synergistic effects of risks, while considering uncertainties, constraints, and the confidence level of evidence and analysis.

These assessments are integrated into the Coastal Resilience Agent-Based Model (CRes-ABM) by simulating the interactions between stakeholders and socioeconomic drivers under various risk scenarios. While climate change research is making significant strides, planning for adaptation and mitigation still needs to be improved. A detailed list of the data sources is provided in Table SM1.

Introducing Custom Future Scenarios by Adapting the Existing CMIP6

Before AR6, a major challenge in climate research was to link RCPs with real-world scenarios for emissions, and political interventions. The SSP and RCP frameworks have offered a structured and standardized foundation for assessing potential climate futures across a range of socioeconomic and emissions scenarios. However, the multifaceted and intricate nature of regional and national contexts demands a more nuanced approach when scrutinizing climate change impacts at sub-global levels. So, for the purpose of this research, a new set of RCPxSSP scenarios have been proposed (Roukounis and Tsihrintzis 2024b), based on the idea of Kok et al. (2019) taking into consideration the characteristics of the case study. More specifically, the scenarios were based on the specific socioeconomic, geopolitical dynamics of the case study area, the environmental variability and the climate change-related policy (Roukounis and Tsihrintzis 2024b).

Scenario 1: GR-SSP2 x RCP4.5 "Symphony of Disintegration"

Under this scenario, Greece has modest development and medium efforts toward sustainability goals. This, alongside the fact that local economy is dependent on tourism activities, will lead to degradation in environmental and public health conditions, despite optimistic expectations. Meanwhile, economic stability is followed by social and capital inequality, higher migration rates, and medium-social cohesion.

Scenario 2: GR-SSP5 x RCP8.5 "For Whom the Storm Tolls" (GR-SSP5)

Under this scenario, Greece has a high rate of growth that is mostly dependent on fossil fuels and is fueled by innovation and competitive markets. Green technologies and sustainability are not priorities for governmental actions. Economy will be less dependent on tourism activities, but opportunities

in the energy and industry sectors will rise. However, high green-house emissions lead to an uncertain future. Climate change actions focus almost exclusively on the economic impact, rather than the environmental. Investments in health and education will increase social cohesion, but will also increase the adoption of energy intensive lifestyles.

Evaluation of Adaptation Measures

Societal infrastructure in the twenty-first century is at risk, due to combined climate change-related hazards, such as SLR, extreme weather conditions, erosion, inundation, and generally the rapid degradation of natural coastal systems (Beck et al. 2018; Hinkel et al. 2013; Hoggart et al. 2014; Hsiang et al. 2017; Reguero et al. 2021). While the most severe impacts of sea-level rise may still be in the future, the need for proactive planning and a reevaluation of development and resource management strategies in coastal areas is immediate. This includes preparing for existing natural hazards like El Niño, hurricanes, and tsunamis.

In the development of our model, a comprehensive evaluation of six distinct types of adaptation measures was conducted using a questionnaire survey addressed to stakeholders (Roukounis and Tsihrintzis 2024a) to assess their effectiveness in mitigating risks associated with sea-level rise (SLR) and coastal flooding. The adaptation measures selected for review are as follows: *community information and participation*, *individual anti-flood measures (property flood-proofing)*, *property insurance*, *nature-based protection measures (living shorelines)*, *hard protection measures*, *managed retreat*, *realignment*, and *setbacks*.

Coastal Resilience Index

Roukounis and Tsihrintzis (2022) reviewed a total of 46 studies that formulate vulnerability and/or resilience indices for climate change related hazards in coastal areas, focusing on physical, social, and economic factors, in an interesting diversity of the developed approaches (e.g., Gornitz 1991; Cutter et al. 2003; Boruff et al. 2005; McGranahan et al. 2007; Tate et al. 2010; Li and Li 2011; Zanetti et al. 2016; Calil et al. 2017; Gargiulo et al. 2020; Zampazas et al. 2022). This led to the development of CResI (Roukounis et al. 2023), a single, unitless aggregated value aiming to assess the resilience of coastal urban and suburban areas to coastal flooding at the local level. The CResI considers 19 variables classified into six categories (i.e., geomorphology, flooding, wave exposition, socioeconomic, land use, infrastructure/functional) as presented in Table SM2. CResI values are from 1.00 (very high resilience) to 5.00 (very low resilience). Within the CResI-ABM, the CResI is used to detect vulnerable points and visualize the effects of adaptation measures on the overall resilience of the coastal area.

Sensitivity Analysis

Sensitivity analysis and Monte Carlo simulations are crucial for CRes-ABM validation process. Through sensitivity analysis, the impact of variance of parameter values is assessed, identifying the most important of them. On the other hand, Monte Carlo simulations offer a probabilistic understanding of how stochastic parameters, under different scenarios, might play out in the real world. Using these together, we ensure that CRes-ABM outcomes are robust, reliable, and able to address real-world cases.

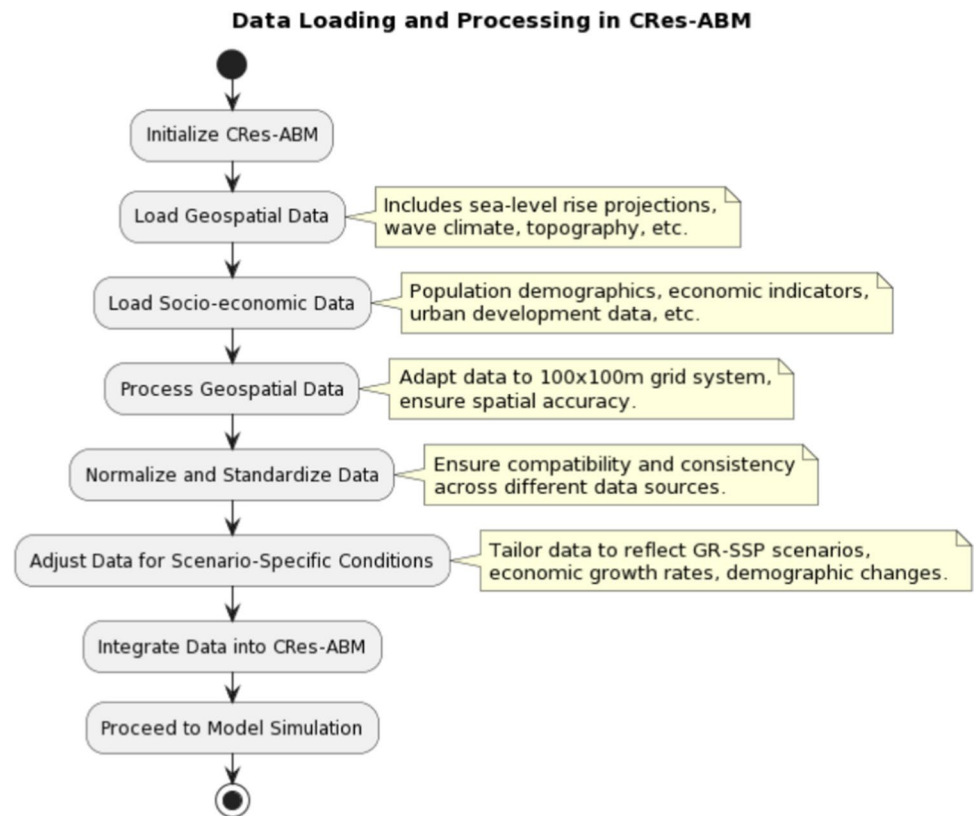
More specifically, the agent's decision thresholds are evaluated, as well as the impact of the stochastic variable of extreme weathers every year, in order to determine their impact in the final outcome. This process helps to identify the most influential factors in CRes-ABM, such as the rate of sea-level rise or the effectiveness of different resilience strategies. By testing a range of input values for each parameter, sensitivity analysis can highlight how changes in environmental conditions or policy decisions might affect CResI.

After understanding the variance in outcomes due to different input parameters, model results are interpreted more accurately. Besides, the GAMA platform offers multiple tools to implement sensitivity analysis within the models, including SOBOL sensitivity analysis (Saltelli 2002), Morris sensitivity analysis (Morris 1991), and beta sensitivity analysis (Baucells and Borgonovo 2013). The GAMA platform's design for batch experiments offers the ability to conduct Monte Carlo simulations within the model, without changing the programming environment or use extra software. In the context of this paper, sensitivity analysis (Saltelli 2002) was implemented. The results of the model are validated through sensitivity analysis and Monte Carlo simulations.

Description of the CRes-ABM

The CRes-ABM is programmed using the GAMA platform simulation software. GAMA (Taillandier et al. 2019) is a powerful and versatile simulation platform designed to provide a user-friendly, dynamic, and comprehensive simulation development environment. Our aim is to address the needs of field experts, modelers, and stakeholders in the creation of spatially explicit multi-agent simulations. As mentioned in the "Agent-Based Modeling Platforms Using GIS Techniques" section, GAMA platform has been equipped with robust capabilities for handling GIS data within its models.

The model incorporates various types of data (Fig. 2), including geospatial information, socio-economic data, and environmental parameters. This data is obtained from reliable and up-to-date databases, such as official government statistics, climate models, and geographic information system (GIS) repositories. As its name suggests, CRes-ABM

Fig. 2 Data input and process outline

integrates data previously used for the development of CResI (Roukounis et al. 2023), as well as future climate and socio-economic projections. We used *geojson* and *json* files to introduce CResI geospatial and socio-economic data to the model and *csv* files to introduce timeseries for climate data. Data insertion and process in the CRes-ABM are outlined in Section SM2.4, while altering parameter interface in GAMA is also shown in Section SM2.4 (Figure SM2).

Last but not least, the normalization and standardization process, necessary for the assessment of the CResI index, is also included in CRes-ABM, to ensure that model's algorithms are consistent across different data sources.

Model Components

The coastal resilience index (CResI), which is essential for comprehending and mitigating the effects of climate change, is used by the CRes-ABM to simulate the environment of coastal urban areas. The coastal resilience index is continuously mapped and visualized by the CResI function in CRes-ABM. First, it provides a baseline for comparison by displaying the resilience state in the absence of any mitigation or adaptation strategies.

Additionally, the model assesses how different resilience measures affect the CResI. This is represented visually in a secondary interface that recalculates and displays a revised CResI (referred to as CResI-ADAPT) that illustrates the

outcomes of adopted adaptation strategies. Both the effects of resilience measures and geomorphological changes are reflected in the dynamic updates of the CResI function. This guarantees that the model accurately represents both the condition of coastal resilience as it exists today and the future efficacy of different approaches. The dual mapping interface offers a thorough understanding of how resilience strategies can change coastal resilience and vulnerability. One shows the adapted state, while the other shows the baseline. Some of the index variables in CRes-ABM fluctuate yearly:

- Wave patterns and sea-level rise: Using time series data, the mean and maximum significant wave heights, as well as the sea-level rise, are updated annually. This guarantees that these shifting environmental factors are accurately reflected in the model.
- Population dynamics: Every 5 years, a specific percentage of each grid cell's population shifts, which has an impact on the CResI score but not the initial number of agents in the cell.

The aggregated data was then applied to every grid cell in our model during the subsequent analysis stage. We scaled each cell's adaptation measure scores from 0 to 1 in order to properly integrate these results. The coastal resilience index (CResI) was then modified using this normalized score (Adaptation Score), yielding an updated CResI_{adapt} value.

Following the implementation of the adaptation measures, this new metric offers a thorough and data-driven depiction of the resilience of the coastal area. This method allows for the exploration of potential future scenarios for coastal resilience planning while also improving the depth of the resilience assessments. The CResI_{adapt} is calculated using Eq. (1):

$$CResI_{adapt} = CResI_{current} - Adaptation_{score} \quad (1)$$

where CResI_{adapt}: resilience index after adaptation measures; CResI_{current}: current resilience index; Adaptation_{score}: normalized score of adaptation measures.

Similar techniques have been applied in numerous studies (e.g., Hallegatte et al. 2012; Balica et al. 2012) to reflect the impacts of interventions by altering indices with fractional values. This method allows for dynamic analysis by treating adaptation measures as changes to the current state. Households and governmental organizations in the coastal environment are represented by individual agents, whose behaviors and interactions are simulated by the agent-based function in CRes-ABM. These agents follow the theory of planned behavior (Ajzen 1991) in their behavior and decision-making, taking into account variables such as risk perception, financial capacity, and policy influences. The model consists of two primary parts and incorporates climate change-related hazards such as storm surges, wave extremes, and sea-level rise.

The CRes-ABM includes climate change-related hazards (e.g., sea-level rise, wave extremes, storm surges) and has two main components:

- Index-based component: Calculates CResI at each time step for updated sea-level rise, wave heights, and population change.
- Agent behavior component: Models agent behavior and the interaction of socio-economic and climate factors for each scenario. Adaptation measures are selected and implemented during this process.

Cell Agents

Three types of agents were considered for this model: Cell agents, containing all the household agents, representing individual households, and one governmental agent, acting as the main decision maker. In future research, a broader set of agents can be considered, including coastal infrastructure and stakeholders from different sectors (industry, tourism, etc.).

For the development of the model, the 100×100 m grid created for the implementation of the CResI was used (Roukounis et al. 2023). For the purpose of CRes-ABM, these cells are separate agents, containing all the geospatial and

socio-economic data previously used for the development of CResI, and are used within the CRes-ABM by household and governmental agents. These data are updated on a yearly basis, according to the description in previous sections, and finally, CResI and CResI_{adapt} are calculated and mapped for each cell agent.

Household Agents

Household agents are placed on the grid, inside each cell agent, represent people living in coastal urban areas. The number of agents in each cell is based on how densely populated the area is, ranging from one to five agents per cell. The number of the agents can be modified, depending on the level of complexity that need to be addressed. These agents are not just placeholders; they actively make decisions that affect the model. Each agent has details like location, income, budget, insurance status, property size, and happiness.

Agents respond to changes in the environment, government policies, and what their neighbors are doing. Their decisions are influenced by their attitudes, what they think others expect them to do, and how much control they feel they have over the situation. Every year, agents decide how much of their income to spend on insurance and flood protection. They consider if they need to buy or renew insurance. Agents who already have insurance are more likely to renew it, while those without insurance have a certain chance of buying it, which can be adjusted in the model. If many neighbors have insurance, an agent is more likely to get or renew insurance. If they cannot afford it, they might seek help from the government. Agents also have a chance of becoming aware of climate change and what they can do to protect themselves, depending on the scenario. They decide on flood-proofing their property based on their income and what their neighbors are doing. If many neighbors have taken flood-proofing measures, they are more likely to do the same.

In the moderate “Symphony of Disintegration” scenario, agents are more cautious about buying insurance due to financial limits and rely more on government help and what others in the community are doing. They are also more likely to need government assistance for flood protection. In the economy and development-friendly “For Whom the Storm Tolls” scenario, agents can more easily afford insurance and flood protection on their own. They are less dependent on government aid and community norms. These agents are also more likely to be aware of climate issues because they have better access to technological advances, information, and resources. The decision-making processes of household agents are shown in detail in Section SM2.5 (Figure SM3 and Figure SM4).

Governmental Agent

The government agent in the model can make decisions that affect the entire area, without having a specific geolocation itself. This agent has detailed knowledge about the environment (characteristics of the grid cell agents), including coastal resilience index (CResI) levels, budget allocations, and sea-level rise data, as well as information about the household agents, such as insurance coverage, and the overall happiness of household agents.

The government agent manages the budget (initially set by the user), which varies by scenario and is renewed annually. Its decisions are based on predefined thresholds for each scenario, set by the authors and experts in the field. Depending on these thresholds, the agent might do nothing, launch community awareness campaigns, enact policies, implement protective measures, or plan for active retreats. The main goal of the government agent is to ensure community safety and well-being of the citizens, and achieve long-term sustainability and economic goals, depending on the priorities for each scenario. It makes its decisions on an annual basis, while constantly updating its strategies based on new data, environmental changes, and feedback from household agents. Governmental decision making is defined by these updating environmental and socio-economic data are grouped in parameters (thresholds). When a threshold is met, different actions are examined (threshold values are shown in Section SM2.4, Table SM3).

In the GR-SSP2xRCP4.5 scenario, the government agent is expected to take a balanced approach, investing in community outreach and education programs to boost public awareness and engagement in climate change mitigation and adaptation strategies. The emphasis is on soft protection measures and supporting small-scale flood protection initiatives that align with the community-focused goals of this scenario.

Conversely, in the GR-SSP5xRCP8.5 scenario, which emphasizes rapid economic growth and technological advancement, the government agent prioritizes infrastructure development and hard protection measures. This involves significant investments in sea walls and major upgrades to address environmental challenges aggressively and immediately. The decision-making process of the government agent is detailed in Section SM2.5 (Figure SM5).

Model Initialization, Data Input, and Threshold Definition

The “model setup” function is essential for starting the simulation. It is activated by pressing the “Start Experiment” button in the user interface. This function places and initializes household agents in the model environment, setting up the simulation. Before running each experiment, users can

change model parameters (Fig. 3). Details on the model variables are presented in Section SM2.4 (Table SM3). A central part of the CRes-ABM model is how agents make decisions. These decisions are influenced by specific thresholds that, when reached, trigger certain actions or policies. Setting these thresholds is an important part of the model setup, based on expert advice and data.

During model development, thresholds were set using existing research and the authors’ expertise, and during calibration, additional expert feedback was collected through a questionnaire survey (Roukounis and Tsihrintzis 2024b).

Results and Discussion

Resilience Mapping

The model produces several outputs. The main advantages of the model are its computational speed, the continuous result visualization, and the elimination of the need to export data for processing in software other than the GAMA platform. The main output while running the software is the generation of a 2D map of agents (cells), showing the CResI value of each cell. Two maps are generated, showing the CResI values under each scenario, one after implementing any adaptation measures, and one without (CResI-Adapt). The comparison is available for each time-step during the simulation. Figures 4 and 5 show the comparison of CResI and CResI_Adapt values for year 2100 and for current/historical conditions.

The overall decrease in CResI values is significant, after the implementation of different adaptation measures. Meanwhile, the vulnerability reduction after adaptation is less clear in scenario GR-SSP5xRCP8.5. Comparing the results for the 2041–2100 projections with the 1976–2005 current/

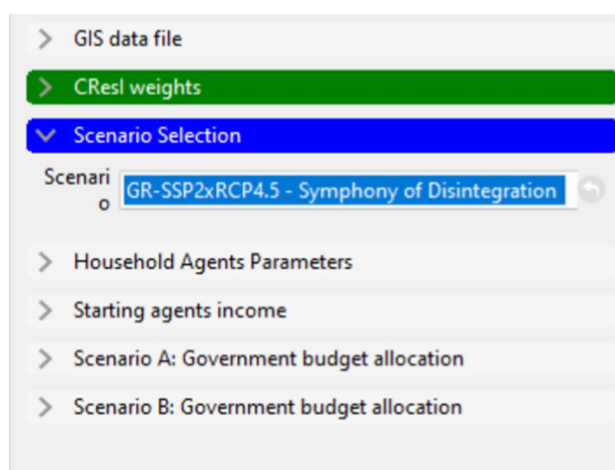
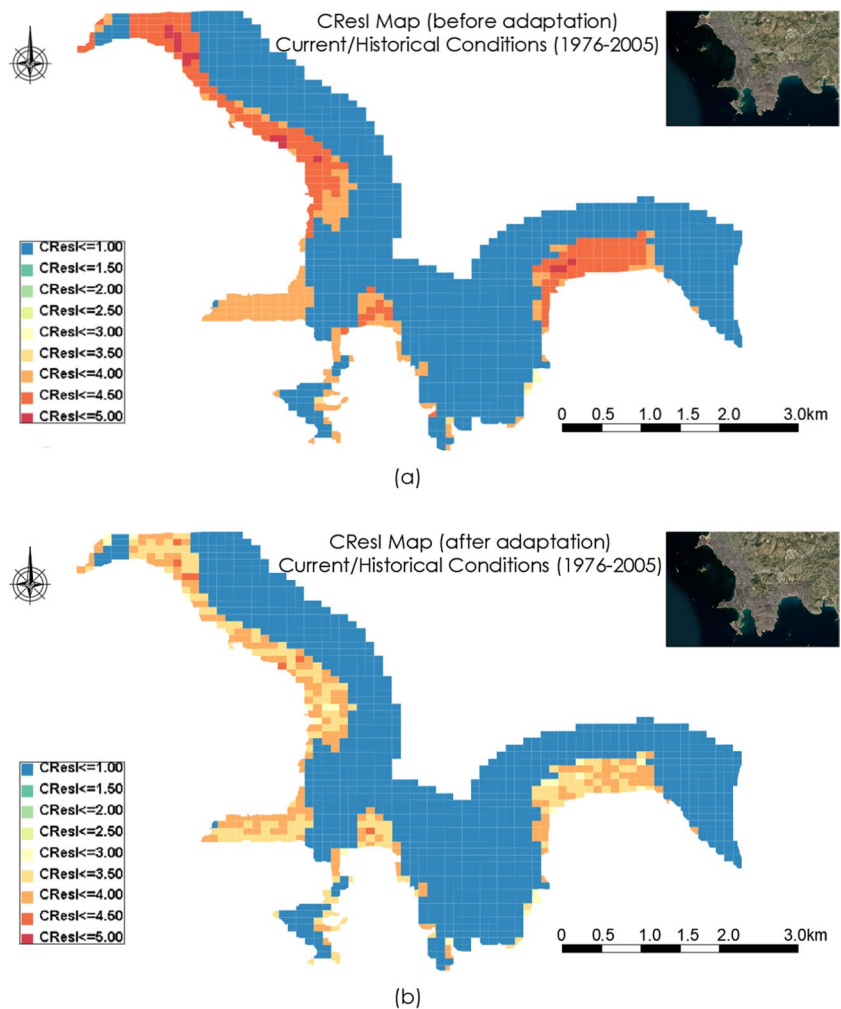


Fig. 3 Selecting initial parameters of the model

Fig. 4 2D Mapping of CResI values without (a) and with (b) the implementation of adaptation measures—current/historical (1976–2005) conditions. Areas in reddish colors have relatively high CResI index and are less resilient to coastal flooding (Snapshot for year 2005)



historical conditions, the model output is consistent for both projections and hindcast 1976–2005. As an output, when running the simulation using the historical data, final result on CResI is between those of Scenario 1 “Symphony of Disintegration” and Scenario 2 “For Whom the Storm Tolls.”

Because of its high cost and the high level of tourism in the area, managed retreat thresholds are never met for this case study. However, the resilience of the coastal region is greatly influenced by household interactions as well as general decisions about insurance, information, and awareness campaigns. In the “For Whom the Storm Tolls” scenario, hard structures are preferred; however, sustainable actions result in a more effective reduction of the CResI index, which implies improved protection of the coastal environment against climate change.

Furthermore, counters and charts are used to monitor the overall number of cells with higher than 4.0 CResI values. In Fig. 6, the number of cells with CResI value greater than 4.0 and 4.5 is shown. It is rather clear that during the winter months of each year, there are more cells with lower resilience levels.

Agents’ Decisions Monitoring

Decisions on insurance and adaptation awareness, as mentioned, are significant for the behavior of the household agents, and therefore, are monitored through every timestep of the model in charts (Fig. 7). By monitoring the development of these variables, we extract useful information about the timing of the decisions of the agents, especially the years when governmental agent decides to implement insurance or awareness policies. For example, in Fig. 7a, the number of informed household agents is generally stable. However, the impact of the implementation of awareness campaigns by the governmental agent in year 18 of the simulation is rather clear.

Meanwhile, in Fig. 7b, it is clear that insurance of households was mandatory 3 times during the simulation. However, if the mandatory measure is not permanent, the number of insured agents drops. Furthermore, we derive significant conclusions on the time of decisions and actions of the governmental agent, as both insurance and awareness status maintain a positive growth rate within the experiment.

Active monitoring of variables in each time step is also made by the monitors in Fig. 8, containing information about both the initialization of the model and the update in each time-step.

Furthermore, the use of hard protection measures and constructions might make household agents feel unhappy, due to the possible environmental and aesthetic degradation of their area. In Fig. 9, the location of unhappy agents is shown (in black color).

As already mentioned, the locations of the unhappy agents are close to areas where the effect of hard protection measures is visible. In Fig. 10, areas affected by hard protection measures are shown in purple, and areas affected by soft protection measures by light blue.

Based on the results presented in Figs. 4, 5, 6, 7, 8, 9, and 10, it is demonstrated that the level of resilience of each region is greatly impacted by the interactions of its agents. Simpler choices play a significant role in determining overall resilience, in addition to the anticipated coping mechanisms such as hard protection measures (breakwaters, etc.). Also,

resilience of a region is greatly influenced by decisions about insurance coverage and awareness initiatives.

Overall, based on social and economic factors, as well as physical data, the behavioral simulation showed various profiles of how people and communities react to natural hazards. This is visualized in the outputs of the model, as in Scenario 2 “For Whom the Storm Tolls,” low-social cohesion leads to resilience based almost exclusively on expensive coastal infrastructure, whereas in Scenario 1 “Symphony of Disintegration,” the decisions of household agents are influenced by their neighbors toward a more sustainable approach.

In Section SM3, additional screenshots of the model environment, including resilience maps and diagrams are presented (Figs. SM6–SM9). Interestingly, comparing these results for random years (2050 and 2070) for both scenarios do not alter the abovementioned conclusions.

Model Calibration

Calibration is a critical process in model development, ensuring that the simulated outcomes closely align with

Fig. 5 2D mapping of CResI values without (a, c) and with (b, d) the implementation of adaptation measures: a, b Scenario 1 “Symphony of Disintegration”; c, d Scenario 2 “For Whom the Storm Tolls” (Snapshot for year 2100). Areas in reddish colors have relatively high CResI index and are less resilient to coastal flooding

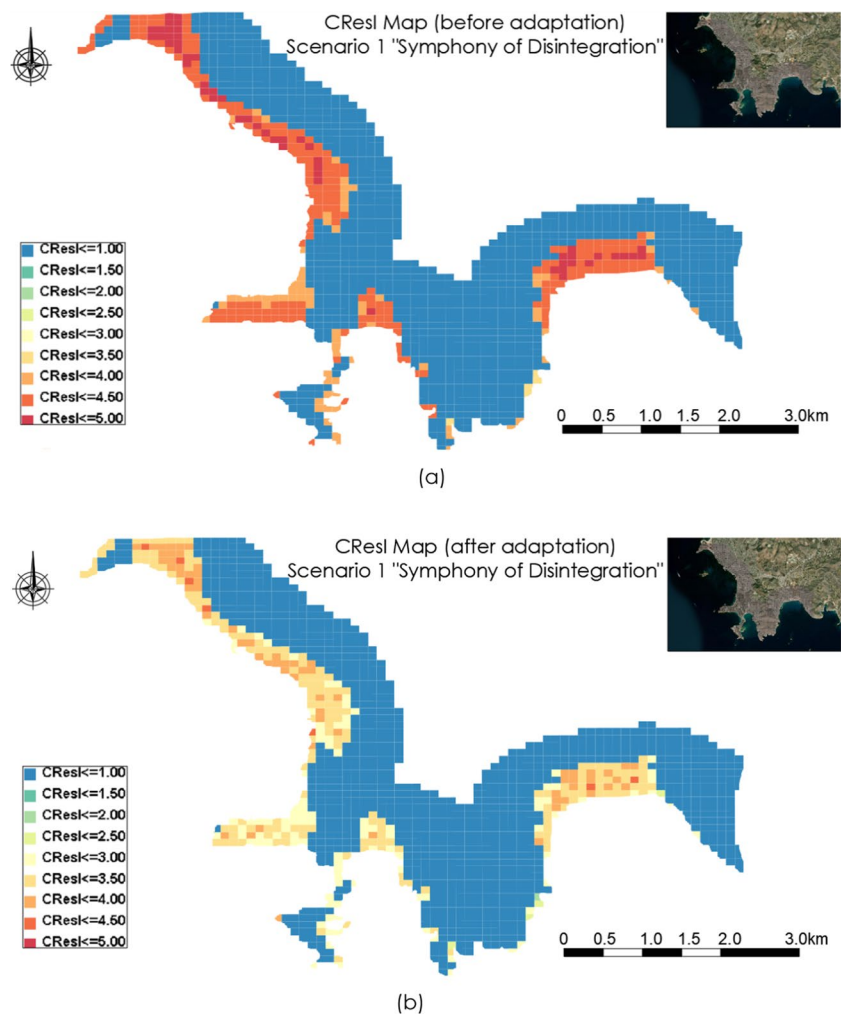
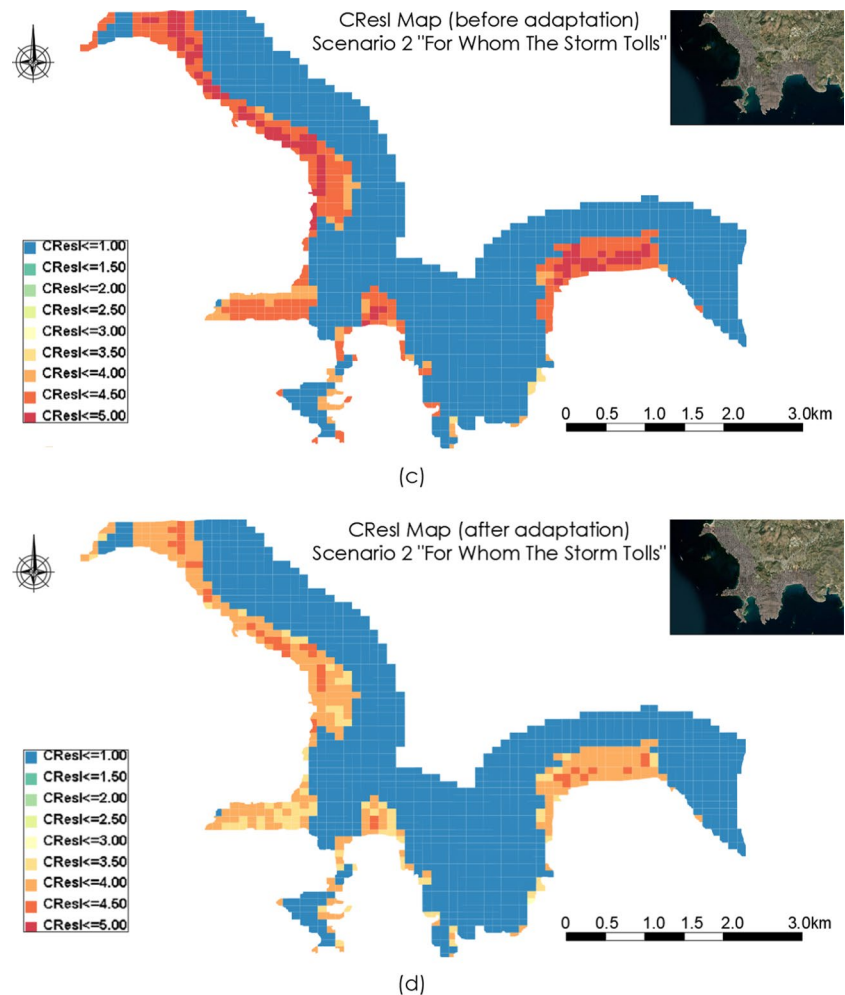


Fig. 5 (continued)



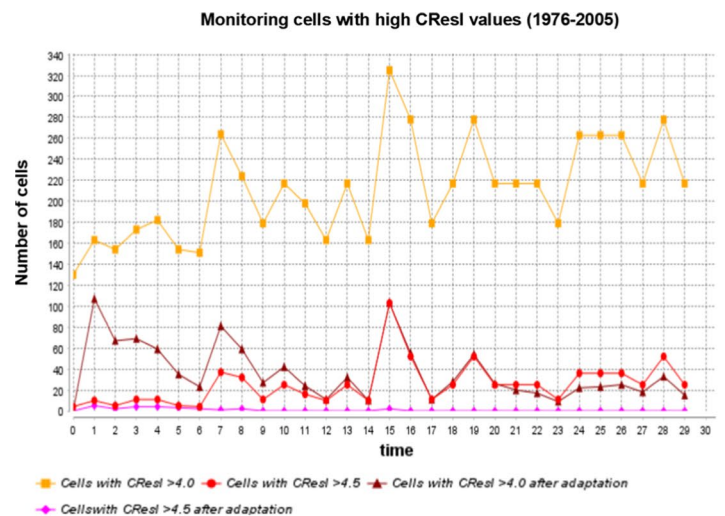
real-world behaviors and observations. For CRes-ABM, calibration plays a pivotal role in accurately representing the diverse and dynamic interactions between stakeholders, infrastructure, and environmental systems. The initial values of the model were selected by the authors, based on the available data, literature, and their expertise in the field. Furthermore, in order to calibrate the model, a questionnaire survey was undertaken, addressed to relevant experts in the field (Roukounis and Tsihrantzis 2024b). The questionnaire was divided into two sections: the first focused on evaluating various adaptation measures, while the second addressed thresholds for implementing these measures. Overall, 47 answers were received from different fields (academia, research institutions, port and marina management, central and local authorities, and relevant consultancies).

Researchers provided specialized knowledge and analytical approaches, while policymakers outlined the regulatory and legislative structures needed to support adaptation initiatives. Field practitioners shared experiences of implementation challenges and successes, offering practical perspectives on what measures are both viable and effective. The survey

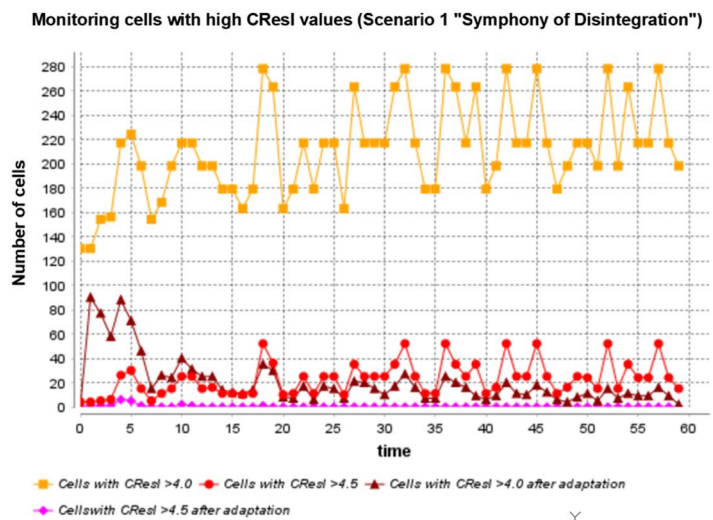
was designed to draw on these diverse experiences, aiming to produce guidance that would inform policy creation and practical decision-making. Contributions from stakeholders helped identify key adaptation priorities, assess the effectiveness of current interventions, and highlight future risks and opportunities. To ensure reliability, the survey results were cross-validated against existing studies and regional adaptation reports.

The questionnaire survey's results showed that different stakeholders have different opinions about how effective adaptation measures are. This diversity makes sense in light of each expert's distinct experiences. For instance, academics are more likely to prioritize environmentally friendly approaches than strategies for economic efficiency. This is especially true for those working in environmental research. Proactive risk reduction strategies were generally ranked lower than community engagement, flood-proofing, and insurance measures by those working in local government and academia. Some groups did not agree with this view. Freelance engineers, on the other hand, score higher on short-term economic efficiency than on long-term

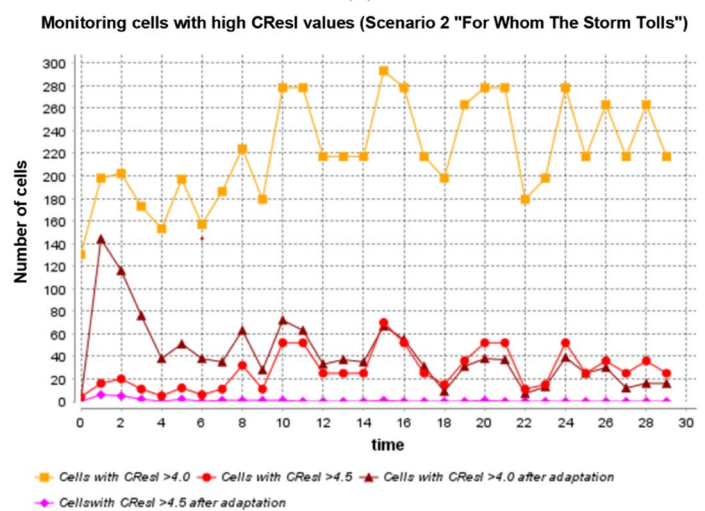
Fig. 6 Number of cells with CResI value higher than 4.0 (time in years) with and without adaptation: **a** current conditions/ historical data (1976–2005); **b** Scenario 1 “Symphony of Disintegration”; **c** Scenario 2 “For Whom the Storm Tolls.” Monitoring is available in every timestep



(a)

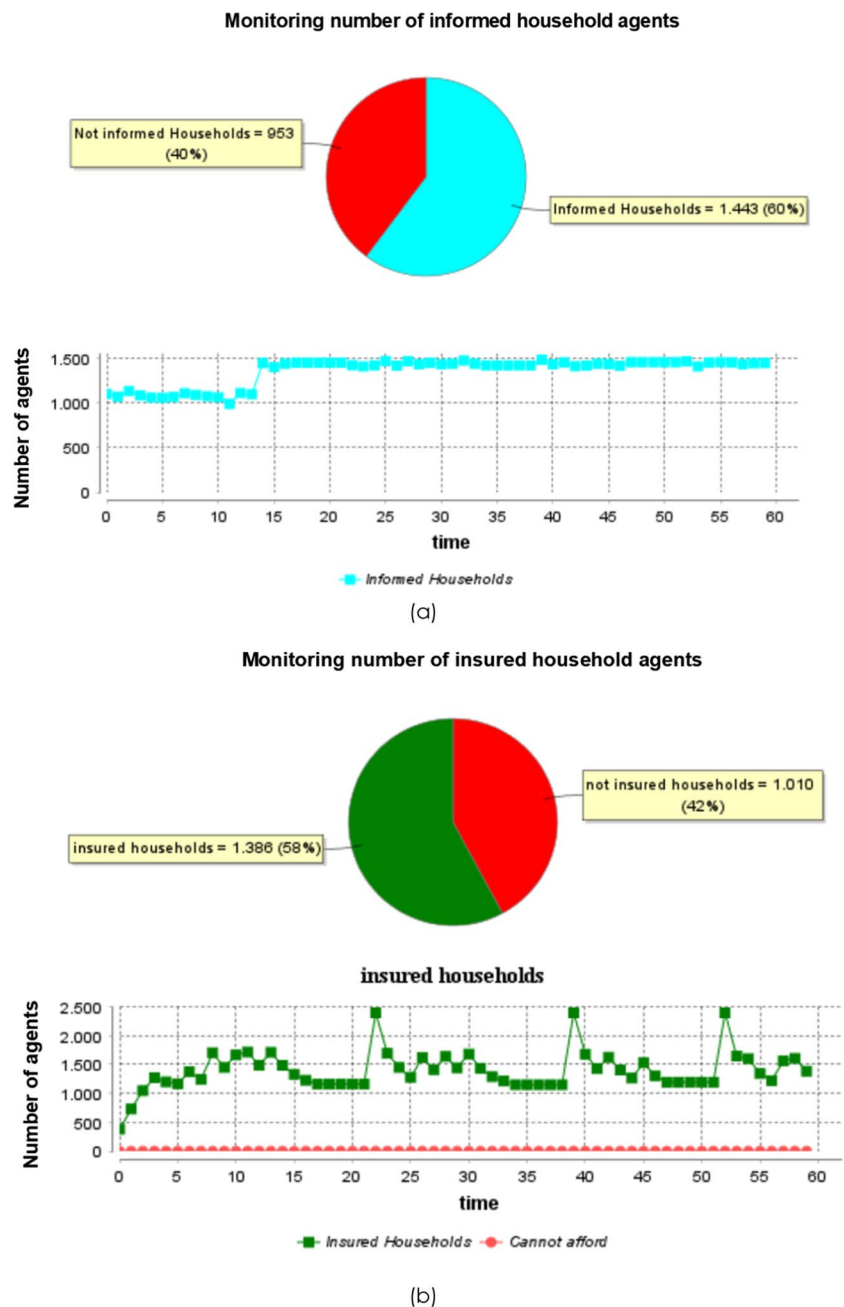


(b)



(c)

Fig. 7 Monitoring the number of informed (a) and insured (b) households (random time-step of Scenario 1 “Symphony of Disintegration”). Monitoring is available in every timestep



sustainability and mitigating climate risk. Together with port authority stakeholders, their interest in nature-based protection measures, however, demonstrates the growing recognition of the importance of ecosystem-based approaches in coastal resilience planning.

However, when determining the adaptation thresholds, comparable variation was noted across the various occupational groups. It is important to note that various groups of related responses were found when part B (adaptation thresholds) and part A (adaptation measures evaluations) questions were analyzed. More

specifically, risk management and infrastructure planning were approached similarly but cautiously by local and port authorities. Academic and research faculty members also underlined the significance of interdisciplinary cooperation and evidence-based decision making. In contrast, the cost-effectiveness profiles of engineering consultants and freelancers indicated more consistent and moderate thresholds. Everything mentioned above emphasizes how difficult it is to make decisions and how important it is for various sectors to work together to develop the best adaptation plan.

Fig. 8 Monitoring variables in the CResI-ABM model

> Number of cells with CResI>4.5:	10	II	X
> Number of cells with CResI>4.0:	163	II	X
> Number of cells with CResI>4.0 after adaptation measures:	15	II	X
> Total number of household agents:	2396	II	X
> Years of insurance assistance activation:	4	II	X
> Number of households that cannot afford insurance:	0	II	X
> Number of households mandated to buy insurance out of budget:	0	II	X
> Cells with hard protection measures:	52	II	X
> Unhappy-unsettled neighborhoods:	11	II	X
> Policy Budget:	3.36E9	II	X
> Soft Protection Measures Budget:	3.3528E9	II	X
> Hard Protection Measures Budget:	3.348E9	II	X
> Active Retreat Budget:	0.0	II	X

Fig. 9 Unhappy/unsettled agents are shown in black color (Snapshot for year 2055)

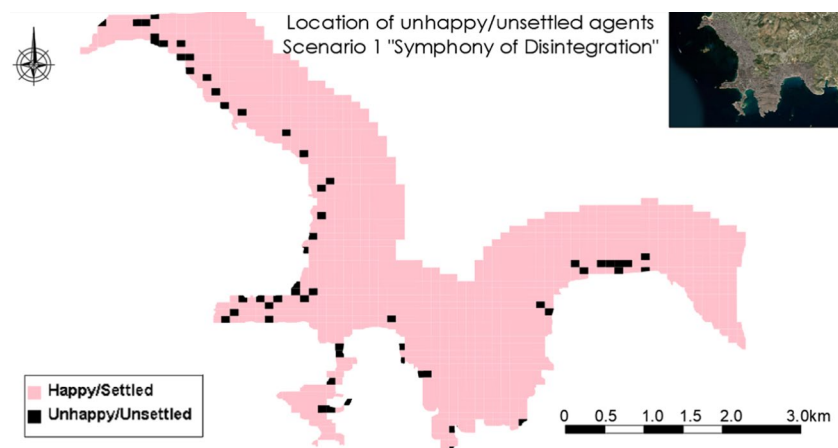


Fig. 10 Areas with soft or hard protection measures. A combination of hard and soft protection is frequently used. Hard protection measures are not always used as the first-choice (Snapshot for year 2085)

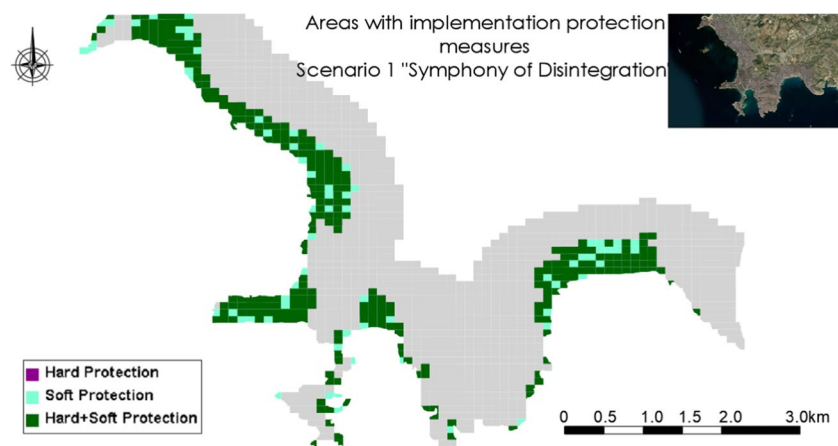


Table 1 Results of BetaD sensitivity analysis after 320 runs

Inputs	Cells with CResI > 4.0	Cells with CResI > 4.5	Cells with CResI_Adapt > 4.0	Cells with CResI_Adapt > 4.5
BETA b Kuiper-based estimator				
SLR threshold	0.06584	0.06584	0.041152	0.005486
Insurance % threshold	0.06995	0.06995	0.091906	0.005486
Awareness % threshold	0.06172	0.06172	0.042524	0.005486
CResI threshold	0.05212	0.05212	0.454046	0.005486
Annual storms	0.06447	0.06447	0.041152	0.005486

Sensitivity Analysis Results

In order to account for randomness and stochastic elements in both environmental factors and human behaviors, CRes-ABM uses Monte Carlo simulations. The accuracy and dependability of CRes-ABM can be evaluated by contrasting the range of results from the Monte Carlo simulations with actual data or forecasts from alternative models. The validation of the model depends on this comparison. This paper applies the exploration of beta distribution analysis (BetaD) (Baucells and Borgonovo 2013). In order to evaluate the impact of a single factor on the entire distribution of outputs, BetaD takes into account all interactions between parameters and is not based on variance in results. BetaD is particularly suited for models with nonlinear interactions, as it captures the complete interdependencies between parameters, rather than focusing solely on individual factor contributions. Simply stated, the beta distribution assesses how much variation on a parameter input will raise the maximum absolute differences. Sensitivity analysis and 320-sample Monte Carlo simulations were used to validate the results of the model. The number of simulations represents about 15% of the population of agents in the model, and therefore, it is capturing the behavior of the overall population. The integrated BetaD algorithm in the GAMA platform is also used to carry out sensitivity analysis. Table 1 displays the results.

This function compares the fitted beta distribution with the empirical distribution of the simulation outputs using Kuiper's statistic. Kuiper's statistic is advantageous for sensitivity analysis because it is particularly sensitive to discrepancies in both the center and tails of the distribution. The difference between the theoretical cumulative distribution function (also known as the CDF) and the empirical cumulative distribution function (ECDF) of the beta distribution is evaluated by Kuiper's statistic. The deviations determined by Kuiper's statistic are used to calculate the sensitivity values, which show how much each input affects the outputs.

The values of Kuiper's statistic, which gauges how well a statistical distribution fits its data, are shown in Table 1. This test is similar to the Kolmogorov–Smirnov test but works

especially well with data that is circular or has wraparound endpoints. As a result, it can be applied to a variety of situations, including goodness-of-fit assessments for periodic distributions. Low sensitivity values are found for “SLR Threshold” across CResI outcomes. This indicates that sea-level rise thresholds have a limited impact on resilience index results within the tested range of conditions. It may suggest that other parameters, such as social or economic factors, play a more substantial role. Testing with a broader range of assumptions could help clarify this relationship.

The CResI is steadily impacted, as evidenced by the moderate sensitivity of insurance and awareness thresholds and the consistent sensitivity of annual storms across a range of outcomes. Therefore, factors like insurance and public awareness affect resilience outcomes, but not as strongly as expected. Finally, as predicted, the CResI threshold exhibits a high sensitivity on the “CResI_adapt” results, showing the importance of targeted adaptation planning, especially when designing measures that respond to specific risks. However, the overall sensitivity values are low, indicating strong model performance. While the sensitivity values are low overall, it is important to recognize that certain model assumptions or unforeseen factors could still introduce variability.

Conclusions

A new innovative agent-based model, the Coastal Resilience Agent-Based Model (CRes-ABM), is presented. It is a computational framework designed to simulate and assess the resilience of coastal areas to the impacts of climate change, including sea-level rise and extreme weather events. It integrates geospatial data, agent-based modeling techniques, and customized climate change scenarios to study how various stakeholders and households within a coastal community interact, adapt, and respond to environmental changes. CRes-ABM aims to inform decision-making and policy development for enhancing the resilience of coastal regions. Although there is a significant number of ABMs in climate change literature, to the best of our

knowledge, there is no other framework available to assess resilience in a dynamic environment. The proposed ABM tries to fill the gap in the literature in this area.

The southwestern waterfront of Attica, Greece, is selected as case study. For this reason, a set of custom scenarios are developed, in order to include the characteristics of the case study area to the mainstream climate change scenarios of AR6. The model consists of the household agents, the governmental agent, and their environment, in a 100×100 m grid, in an accurate representation of reality. A set of adaptation measures is proposed, based on the relevant literature and the authors' expertise, and its impact is evaluated by experts through a questionnaire survey (Roukounis and Tsihrintzis 2024) and added in the model. The model is developed within the GAMA platform using statistical, temporal, spatial, and descriptive data, previously edited using GIS techniques.

The originality of the methodology lies in the fact that in the existing literature this combination of methods has not been applied so far to address coastal flooding due to climate change. As a result, resilience maps emerged for both scenarios, based on the CResI index. As expected in the hypothesis, the "For Whom the Storm Tolls" GR-SSP5xRCP8.5 scenario is the most challenging, as fossil fuel development continues to grow and adaptation to coastal flooding relies on reactive measures or proactive measures with low sustainability profile (like quay walls, extensive breakwaters etc.). Managed retreat thresholds are not met for this case study under no circumstances, mainly due to its high cost and advanced touristic activity in the area. On the other hand, the interactions of households and the overall decisions on insurance, information, and awareness campaigns play a significant role on the resilience of the coastal area. Hard structures are preferred in Scenario 2 "For Whom the Storm Tolls"; however, sustainable actions, lead to more effective reduction of the CResI index. The results of the model are validated through beta sensitivity analysis and Monte Carlo simulations, using a 200-iteration approach. Sensitivity values are low for all the variables under consideration, signifying robust results from the model.

To conclude, this research demonstrates the value and the potential of ABM resilience models. The model indicates the possibilities raised from the synergy of multidisciplinary research of complex human–environment systems. The proposed framework will be the base for other case studies, and expansions adding more types of agents, to reflect the high-detailed complexities of the real world, contributing significantly to the research world.

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Data Availability The data presented in this study are available upon reasonable request from the corresponding author.

Declarations

Conflict of Interest The authors declare no competing interests.

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